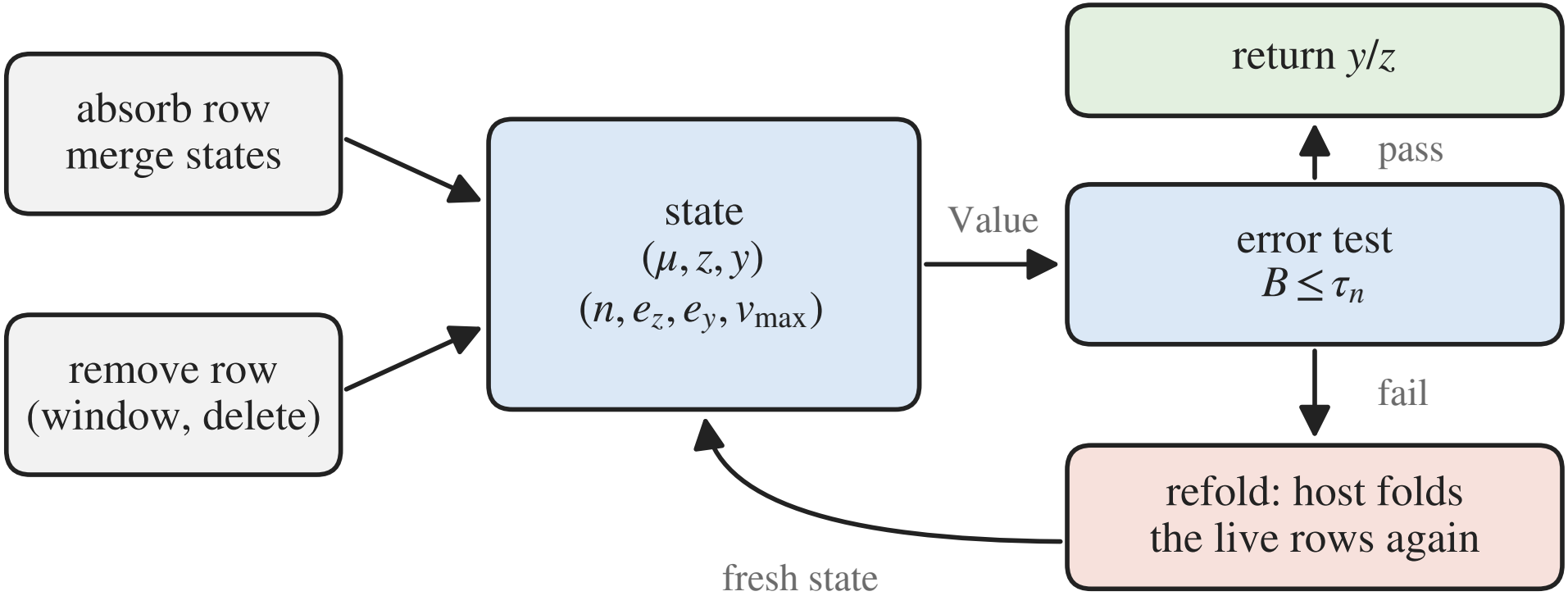

constant size, $O(1)$ arithmetic per update;
only a failed test costs a pass over the live rows